

The Second Law of Thermodynamics

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1 Second Law of Thermodynamics

The **First Law of Thermodynamics** expresses the conservation of energy: energy can neither be created nor destroyed, but only transformed from one form into another. While this law constrains the total energy balance of a process, it places *no restriction on the direction* in which a process may occur. In particular, it does not distinguish between physically realizable processes and those that are never observed in nature.

The **Second Law of Thermodynamics** introduces a fundamental asymmetry in natural processes. It determines the *directionality* of spontaneous transformations, explains the observed irreversibility of macroscopic phenomena, and imposes intrinsic limits on the efficiency of energy conversion, especially in heat engines and refrigerators.

Historically, the Second Law has been formulated in several equivalent ways, each emphasizing a different physical aspect. Two of the most commonly cited formulations are:

- **Clausius statement:** *No process is possible whose sole result is the transfer of heat from a colder body to a hotter body without the input of external work.*
- **Kelvin–Planck statement:** *No process is possible whose sole result is the absorption of heat from a single thermal reservoir and its complete conversion into work.*

The Clausius statement highlights the impossibility of spontaneous heat flow against a temperature gradient, whereas the Kelvin–Planck statement rules out the existence of a perfect heat engine operating with 100% efficiency.

Despite their seemingly different emphases, these two statements are *logically equivalent*: a hypothetical violation of one would necessarily imply a violation of the other. In the following section, we present a detailed and constructive proof of this equivalence, thereby demonstrating that both formulations express the same fundamental physical principle.

2 Equivalence of Clausius and Kelvin–Planck Statements

We now prove their equivalence by contradiction.

2.1 Clausius Violation $\implies$ Kelvin–Planck Violation

Assumption: There exists a device (Device A) that transfers heat Q from a cold reservoir at temperature T_C to a hot reservoir at $T_H > T_C$ *without any work input*.

Device A (Clausius–violating pump):

$$W_A = 0, \quad Q_C^{(A)} = -Q, \quad Q_H^{(A)} = +Q,$$

where $Q_C^{(A)}$ is heat lost by the cold reservoir and $Q_H^{(A)}$ is heat gained by the hot reservoir.

Device B (ordinary heat engine):

$$W_B = Q_H^{(B)} - Q_C^{(B)}, \quad Q_H^{(B)} > Q_C^{(B)},$$

absorbing $Q_H^{(B)}$ from the hot reservoir and rejecting $Q_C^{(B)}$ to the cold reservoir.

Device A: Heat transfer from cold to hot without work

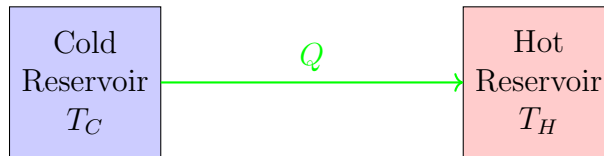


Figure 1: Device A transfers heat spontaneously from cold to hot reservoir.

Now consider a heat engine (Device B) operating between the same two reservoirs:

Device B: Heat engine producing work

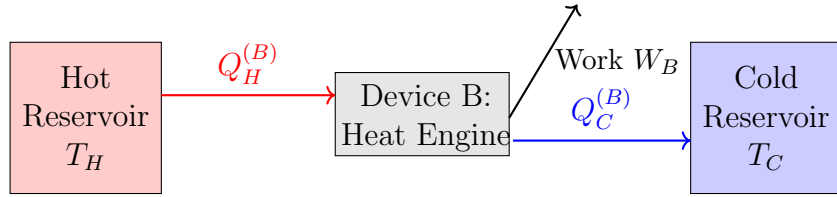


Figure 2: Device B absorbs heat from the hot reservoir, converts part of it into work, and rejects the remainder to the cold reservoir.

Combined operation: The net changes in the reservoirs are:

$$\begin{aligned}\Delta Q_H &= Q_H^{(A)} - Q_H^{(B)} = Q - Q_H^{(B)}, \\ \Delta Q_C &= Q_C^{(A)} + Q_C^{(B)} = (-Q) + Q_C^{(B)}.\end{aligned}$$

Choose $Q = Q_C^{(B)}$. Then:

$$\begin{aligned}\Delta Q_C &= -Q + Q = 0 \quad (\text{cold reservoir unchanged}), \\ \Delta Q_H &= Q_C^{(B)} - Q_H^{(B)} = -(Q_H^{(B)} - Q_C^{(B)}) = -W_B.\end{aligned}$$

Thus the hot reservoir loses W_B , which appears entirely as work output:

$$Q_{\text{from hot}} = W_{\text{out}}.$$

The *sole effect* is complete conversion of heat from a *single reservoir* into work, violating Kelvin–Planck.

2.2 Kelvin–Planck Violation $\implies$ Clausius Violation

Assumption: There exists a device (Device C) that converts heat Q from a single reservoir at T_H completely into work:

$$W_C = Q, \quad Q_{\text{rejected}} = 0.$$

Net effect on the hot reservoir:

$$\Delta Q_{\text{hot from C}} = -Q.$$

Device C: Perfect heat engine

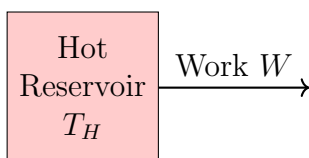


Figure 3: Device C converts heat fully into work (impossible by Kelvin–Planck).

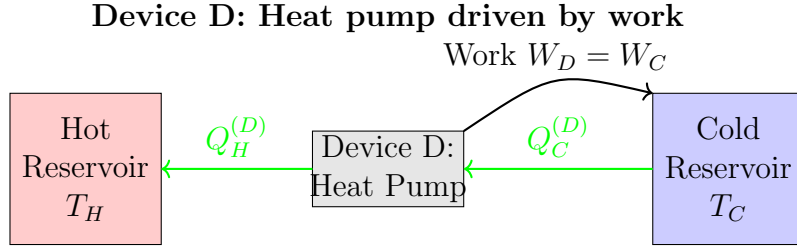


Figure 4: Device D uses work to transfer heat from cold to hot reservoir.

Device D (heat pump): Device D is driven by the work output of Device C and uses it to transfer heat from the cold reservoir to the hot reservoir. For an ideal operation,

$$W_D + Q_C^{(D)} = Q_H^{(D)},$$

where $W_D = Q$ is the work supplied by Device C. Hence,

$$Q + Q_C^{(D)} = Q_H^{(D)}.$$

The total change in heat for the hot reservoir is then

$$\Delta Q_{\text{hot total}} = (-Q) + Q_H^{(D)} = Q_C^{(D)}.$$

For the cold reservoir,

$$\Delta Q_{\text{cold total}} = -Q_C^{(D)}.$$

Therefore, the *net effect* of operating Devices C and D together is a transfer of heat $Q_C^{(D)}$ from the cold reservoir to the hot reservoir, **without any net work exchange** with the surroundings. This spontaneous heat flow from cold to hot constitutes a **violation of the Clausius statement** of the second law of thermodynamics.

Conclusion

A violation of either statement necessarily implies a violation of the other. Therefore, the Clausius and Kelvin–Planck statements are logically equivalent.

3 The Carnot Engine: An Ideal Heat Engine

The Carnot engine is a fundamental concept in thermodynamics that defines the theoretical maximum efficiency any heat engine can achieve when operating between two thermal reservoirs. It is an idealized engine introduced by Sadi Carnot in 1824, and although no real engine can achieve its ideal operation, the Carnot engine provides an essential benchmark for the performance of all heat engines.

3.1 Physical Setup

The Carnot engine operates reversibly between two thermal reservoirs:

- A **hot reservoir** maintained at a constant temperature T_H , from which the engine absorbs heat.
- A **cold reservoir** maintained at a constant temperature $T_C < T_H$, to which the engine rejects heat.

The engine absorbs heat Q_H from the hot reservoir, converts part of it into work W , and rejects the remaining heat Q_C to the cold reservoir.

$$Q_H = W + Q_C,$$

where

$$Q_H > 0, \quad W > 0, \quad Q_C > 0.$$

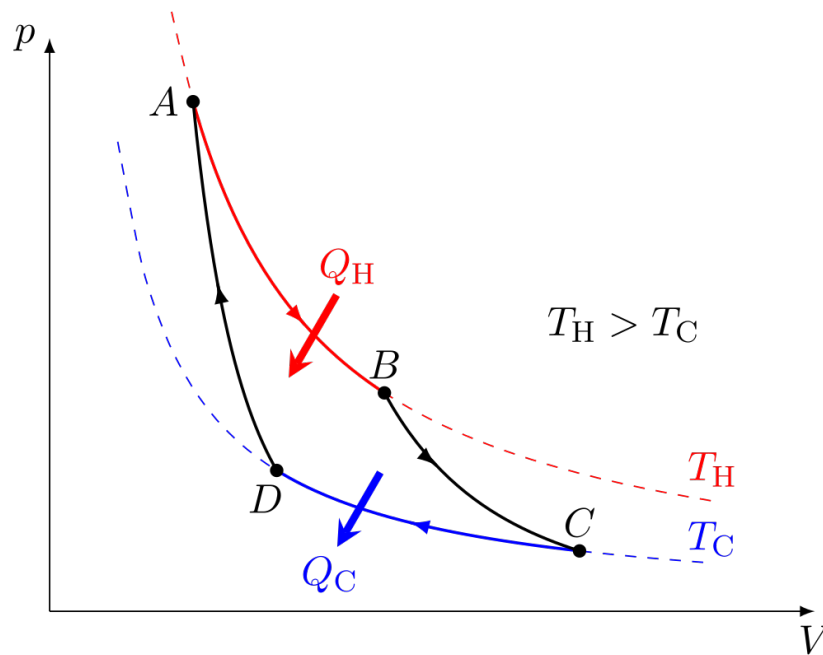


Figure 5: Carnot cycle P - V diagram. Image from Wikipedia: “Carnot heat engine”.

3.2 Reversible Carnot Cycle

The Carnot engine undergoes a **reversible cyclic process**, called the **Carnot cycle**, consisting of four idealized thermodynamic steps. Each step is driven either by a temperature difference or by mechanical work, and reversibility requires that all processes occur quasi-statically.

1. Isothermal Expansion (at T_H):

The working substance is placed in thermal contact with a hot reservoir at temperature T_H . Because the gas and the reservoir have the same temperature, heat transfer occurs without a finite temperature gradient, ensuring reversibility.

As heat Q_H flows into the gas from the hot reservoir, the gas tends to increase its internal energy. However, since the temperature is held constant, this increase is exactly compensated by the gas doing work on the surroundings. Microscopically, the absorbed heat maintains the average kinetic energy of the molecules while allowing the gas to expand and push the piston outward.

Thus,

$$\Delta U = 0, \quad Q_H = W_{\text{exp}}.$$

The absorption of heat is therefore the *cause* of the expansion: the heat input supplies the energy required for the gas to perform work.

2. Adiabatic Expansion:

After the isothermal expansion, the gas is thermally insulated from the reservoirs, so that no heat exchange is possible:

$$Q = 0.$$

The gas continues to expand due to its internal pressure and does work on the surroundings. Since no heat enters the system, this work must come entirely from the internal energy of the gas. As a result, the internal energy decreases, and the temperature drops from T_H to T_C .

This explains why cooling occurs during adiabatic expansion: expansion without heat input necessarily reduces the internal energy.

3. Isothermal Compression (at T_C):

The gas is now placed in thermal contact with a cold reservoir at temperature T_C . The surroundings compress the gas slowly, doing work on it. Compression tends to increase the internal energy, but because the temperature is kept constant, this excess energy must be removed as heat.

Consequently, heat Q_C flows from the gas into the cold reservoir. The rejection of heat ensures that the average molecular kinetic energy remains fixed at the value corresponding to T_C .

Thus,

$$\Delta U = 0, \quad W_{\text{comp}} = Q_C.$$

Heat release is therefore the *result* of compression at constant temperature.

4. Adiabatic Compression:

Finally, the gas is again thermally insulated and compressed further. Since no heat exchange is allowed, all the work done on the gas increases its internal energy. This increase raises the temperature from T_C back to T_H , returning the gas to its original thermodynamic state.

This step prepares the system to repeat the cycle and explains why heating occurs during adiabatic compression: work done on the gas must appear as an increase in internal energy.

3.3 Key Assumptions

The ideal performance of the Carnot engine relies on the following fundamental assumptions:

- **Reversibility:** Each process is carried out quasi-statically, so the system passes through a continuous sequence of equilibrium states. There is no entropy production within the engine or at its boundaries, which guarantees that the cycle achieves the maximum possible efficiency allowed by the Second Law of Thermodynamics.
- **Absence of dissipation:** The piston and cylinder are assumed to be perfectly frictionless, and all thermal contacts are ideal. No mechanical energy is lost to friction, viscosity, or turbulence, and no heat leaks occur except during the intended isothermal steps.
- **Infinite thermal reservoirs:** The hot and cold reservoirs are assumed to be so large that the exchange of heat does not change their temperatures. This ensures that the isothermal processes occur at fixed temperatures T_H and T_C , respectively.

3.4 Significance of the Carnot Engine

- **Maximum Efficiency:** The Carnot engine sets the upper limit for the efficiency of any heat engine operating between temperatures T_H and T_C .
- **Universality:** Its efficiency depends only on the temperatures of the reservoirs, independent of the working substance or engine design.
- **Foundation for Thermodynamics:** The Carnot cycle was instrumental in formulating the Second Law of Thermodynamics and the concept of entropy.

3.5 Interpretation of the Reverse Process

The *reverse* of a cyclic process $\mathcal{O}$ is defined as the same sequence of thermodynamic states traversed in the opposite direction, corresponding to running the cycle backward in time. All heat and work exchanges therefore occur with their signs reversed.

For a heat engine operating cyclically between two heat reservoirs, reversing the cycle implies that the device no longer converts heat into net work. Instead, external work must be supplied to the system in order to transfer heat from the colder reservoir to the hotter one. In this mode of operation, the device functions as a refrigerator or a heat pump.

More precisely, the reverse cycle:

- Absorbs heat from the cold reservoir and rejects heat to the hot reservoir,

- Requires an input of mechanical work to drive this heat transfer,
- Remains a cyclic process, returning the system to its initial thermodynamic state after each cycle, while retracing the original path in state space in the opposite direction.

Interpreting the reverse process in this manner is essential for the proof of the Clausius theorem, as it allows forward and reverse cycles to be compared on equal footing under the assumption of reversibility.

3.6 Efficiency of the Carnot Engine

Over one complete Carnot cycle, the working substance performs work on the environment during the expansion processes and has work performed on it during the compression processes. Since the system returns to its initial state after one cycle, its internal energy undergoes no net change, $\Delta U = 0$. According to the First Law of Thermodynamics, the net work output is therefore

$$W = Q_H - Q_C,$$

where Q_H is the heat absorbed from the hot reservoir and Q_C is the heat rejected to the cold reservoir.

The efficiency η of a heat engine is defined as the ratio of the net work output to the heat absorbed from the hot reservoir:

$$\eta := \frac{W}{Q_H},$$

which immediately gives

$$\eta = 1 - \frac{Q_C}{Q_H}.$$

In a later section, we will show that for a reversible Carnot cycle the ratio Q_C/Q_H depends only on the temperatures of the two reservoirs, leading to the celebrated Carnot efficiency formula. For the present discussion, it is sufficient to note that $Q_C < Q_H$, and therefore $\eta < 1$: no heat engine can convert all of the absorbed heat into work.

3.7 Practical Considerations

While the Carnot engine is an idealized concept and cannot be realized perfectly due to irreversibilities such as friction, heat losses, and finite-time processes, it provides a crucial theoretical benchmark. Real engines aim to approach Carnot efficiency by reducing these sources of loss and operating as closely as possible to reversible conditions.

The Carnot engine clearly demonstrates that the Second Law of Thermodynamics places an upper limit on the efficiency of all heat engines operating between two reservoirs. This limit depends only on the temperatures of the reservoirs, not on the working substance or the details of the engine's design.

3.8 Carnot's Theorem

Carnot's theorem states:

Theorem 1 (Carnot's Theorem). *No heat engine operating between two heat reservoirs can be more efficient than a reversible Carnot engine operating between the same reservoirs.*

Proof.

Definitions of Heat Quantities

Consider two heat engines operating between the same hot and cold reservoirs:

- Engine E (the hypothetical engine more efficient than the Carnot engine) absorbs heat $Q_H^{(E)}$ from the hot reservoir, rejects heat $Q_C^{(E)}$ to the cold reservoir, and produces work $W_E = Q_H^{(E)} - Q_C^{(E)}$.
- Engine C (the **reversible** Carnot engine) absorbs heat $Q_H^{(C)}$ from the hot reservoir and rejects heat $Q_C^{(C)}$ to the cold reservoir. These quantities are adjusted so that the work output of C , $W_C = Q_H^{(C)} - Q_C^{(C)}$, is equal to that of engine E , i.e., $W_C = W_E$.

Now, we operate engine E forward as a heat engine, producing work W_E , and simultaneously run engine C in reverse as a refrigerator. Since C is a *reversible Carnot engine*, it uses the same amount of work, $W_C = W_E$, to transfer heat from the cold reservoir to the hot reservoir.

Net Heat and Work Exchanges

- The **hot reservoir** loses heat $Q_H^{(E)}$ to engine E and gains heat $Q_H^{(C)}$ from refrigerator C . Therefore, the net heat change of the hot reservoir is:

$$Q_H^{(\text{net})} = -Q_H^{(E)} + Q_H^{(C)}.$$

- The **cold reservoir** loses heat $Q_C^{(C)}$ to refrigerator C and gains heat $Q_C^{(E)}$ from engine E . So the net heat change of the cold reservoir is:

$$Q_C^{(\text{net})} = -Q_C^{(C)} + Q_C^{(E)}.$$

- The net work done on the surroundings is zero because the work produced by engine E powers the refrigerator C :

$$W_{\text{net}} = W_E - W_C = 0.$$

Contradiction

Since engine E is assumed to be more efficient than the reversible Carnot engine C , it follows that engine E produces more work per unit heat absorbed from the hot reservoir:

$$\frac{W_E}{Q_H^{(E)}} > \frac{W_C}{Q_H^{(C)}}.$$

We run engine E forward and engine C in reverse (as a refrigerator), using the work output of E to drive C , so

$$W_E = W_C = W.$$

By definition of work:

$$W = Q_H^{(E)} - Q_C^{(E)} = Q_H^{(C)} - Q_C^{(C)}.$$

From the efficiency inequality and equal work, it follows that

$$Q_H^{(E)} < Q_H^{(C)}.$$

Using the work expressions to solve for the heat rejected/absorbed at the cold reservoir,

$$Q_C^{(E)} = Q_H^{(E)} - W, \quad Q_C^{(C)} = Q_H^{(C)} - W,$$

which implies

$$Q_C^{(E)} < Q_C^{(C)}.$$

Therefore, the net heat exchanged with the reservoirs is

$$Q_H^{(\text{net})} = Q_H^{(C)} - Q_H^{(E)} > 0, \quad Q_C^{(\text{net})} = Q_C^{(E)} - Q_C^{(C)} < 0.$$

Moreover, the heat gained by the hot reservoir equals the magnitude of the heat lost by the cold reservoir,

$$Q_H^{(\text{net})} = |Q_C^{(\text{net})}|.$$

Thus, heat flows spontaneously from the cold reservoir to the hot reservoir without any net work exchange, in direct violation of the Clausius statement of the Second Law.

Therefore, the initial assumption that engine E is more efficient than the Carnot engine C must be false.

□

Carnot's theorem establishes the Carnot cycle efficiency as the theoretical upper bound for all heat engines, and this result underpins the absolute temperature scale and the concept of entropy in thermodynamics. From Carnot's theorem, the following fundamental result is obtained immediately:

Corollary 1. *All reversible Carnot heat engines operating between the same two reservoirs have identical efficiency.*

According to **Carnot's corollary**, for any reversible heat engine operating between two reservoirs, the ratio of heats exchanged satisfies a universal relation:

$$\frac{Q_1}{Q_2} = f(\theta_2, \theta_1), \quad (1)$$

where f is a universal function, and θ_1 and θ_2 are parameters that uniquely characterize the two reservoirs. This relation holds independently of the working substance or the details of the engine, reflecting the fundamental universality of reversible processes.

4 Definition of Absolute Temperature

Carnot's theorem tells us that all reversible heat engines operating between the same two reservoirs have the **same efficiency**, regardless of their working substances or design. This universality implies a fundamental relationship among the heat exchanged by reversible engines.

To reveal this relationship, consider a system of **three thermal reservoirs** at different temperatures T_1, T_2, T_3 , and reversible engines operating pairwise between them.

4.1 Setup: Three Reservoirs and Reversible Engines

- **Engine A** runs between reservoirs 1 and 2, absorbing heat Q_1 from reservoir 1 and rejecting heat Q_2 to reservoir 2.
- **Engine B** runs between reservoirs 2 and 3, absorbing heat Q_2 from reservoir 2 and rejecting heat Q_3 to reservoir 3.
- **Engine C** runs directly between reservoirs 1 and 3, absorbing heat Q_1 from reservoir 1 and rejecting heat Q_3 to reservoir 3.

Each engine is reversible, so its efficiency depends only on the pair of reservoirs it connects, not on its construction or working substance.

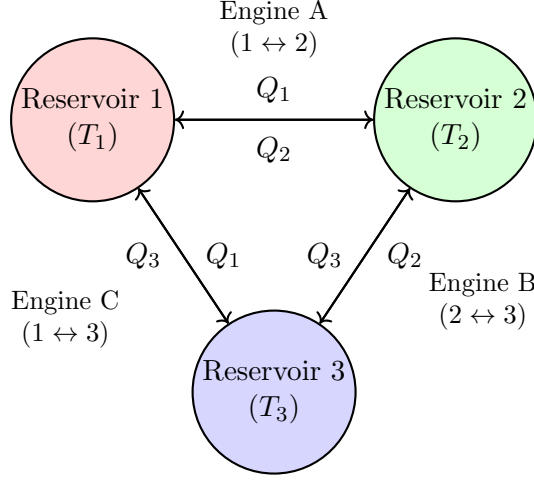


Figure 6: Three reservoirs and reversible engines operating pairwise between them.

4.2 Defining Absolute Temperature via the Multiplicative Property of Heat Ratios

Since engines A and B are reversible heat engines, we can **operate them in sequence**: engine A functions as a heat engine between reservoirs 1 and 2, followed by engine B as a heat engine between reservoirs 2 and 3. The combined operation then effectively behaves as a single reversible heat engine operating directly between reservoirs 1 and 3. Algebraically, the sequential operation satisfies the simple relation

$$\frac{Q_3}{Q_1} = \frac{Q_2}{Q_1} \times \frac{Q_3}{Q_2},$$

which, according to Eq. (1), leads to

$$f(\theta_1, \theta_3) = f(\theta_1, \theta_2) f(\theta_2, \theta_3).$$

This equation serves as the defining relation that determines the functional form of f . To solve this functional equation, let us fix $\theta_3 = \theta_0$. Then

$$f(\theta_1, \theta_2) = \frac{f(\theta_1, \theta_0)}{f(\theta_2, \theta_0)}.$$

Define

$$g(\theta) := [f(\theta, \theta_0)]^{-1},$$

so that

$$f(\theta_1, \theta_2) = \frac{g(\theta_2)}{g(\theta_1)}.$$

We now introduce the **absolute thermodynamic temperature**, defined to be proportional to $g(\theta)$. Choosing the proportionality constant as unity gives

$$T = g(\theta),$$

and therefore

$$\frac{Q_1}{Q_2} = f(\theta_2, \theta_1) = \frac{T_1}{T_2}.$$

Hence, the efficiency η of any reversible engine operating between two reservoirs at temperatures T_H and T_C is

$$\eta = 1 - \frac{T_C}{T_H}$$

Remarks

- This absolute temperature scale arises purely from the **universality and multiplicative property of reversible engine efficiencies**; it does not depend on any specific thermometer or material.
- The choice of the constant factor (units) is conventional—for example, fixing the triple point of water to 273.16 units defines the Kelvin scale.
- This construction relies only on heat/work bookkeeping and the properties of reversible processes—**no use of entropy or other thermodynamic quantities is needed at this stage**.

5 Heat Engines vs. Refrigerators: Definitions, Principles, and Efficiency

Both **heat engines** and **refrigeration devices** (including **refrigerators** and **heat pumps**) operate on the fundamental principles of thermodynamics, particularly the **Second Law of Thermodynamics**. All these devices involve the transfer of heat between two thermal reservoirs and the conversion of energy between heat and work.

A **heat engine** absorbs heat Q_H from a high-temperature reservoir, converts part of this energy into useful work W , and rejects the remaining heat Q_C to a low-temperature reservoir. Its primary purpose is to **produce work** from a temperature difference.

In contrast, a **refrigerator** and a **heat pump** operate in the **reverse direction**. By consuming external work W , they transfer heat from a **cold reservoir** to a **hot reservoir**, against the natural direction of spontaneous heat flow. The distinction between the two lies in their intended function: a refrigerator's goal is to **remove heat** from the cold space to keep it cool, whereas a heat pump aims to **deliver heat** to the hot space to keep it warm.

Thus, these three devices represent different modes of energy conversion governed by the same thermodynamic laws, differing only in their **direction of operation** and **intended purpose**.

5.1 Heat Engine

A heat engine is a device that converts heat energy into work by operating in a cyclic process.

Working Principle A heat engine extracts heat from a high-temperature reservoir at temperature T_H , converts part of it into work, and rejects the remaining heat to a low-temperature reservoir at temperature T_C .

Key Equations Energy Conservation (First Law of Thermodynamics):

$$W = Q_H - Q_C,$$

where

- Q_H : heat absorbed from the hot reservoir at temperature T_H ,
- Q_C : heat rejected to the cold reservoir at temperature T_C ,
- W : useful work output.

Thermal Efficiency η :

$$\eta = \frac{W}{Q_H} = 1 - \frac{Q_C}{Q_H}.$$

For an ideal (Carnot) engine operating reversibly between two reservoirs,

$$\frac{Q_C}{Q_H} = \frac{T_C}{T_H},$$

so the maximum efficiency is

$$\eta_{\text{Carnot}} = 1 - \frac{T_C}{T_H}.$$

Since $T_C > 0$, the efficiency is always less than 100%, consistent with the Kelvin statement of the Second Law.

Examples of Heat Engines

- Steam engines
- Internal combustion engines (car engines)
- Power plants (thermal, nuclear)

5.2 Refrigerator (or Heat Pump)

A refrigerator is a device that transfers heat from a cold reservoir to a hot reservoir by using external work. This operation is essentially the reverse of a heat engine.

Working Principle A refrigerator extracts heat Q_C from a low-temperature reservoir at temperature T_C and expels heat Q_H to a high-temperature reservoir at temperature T_H by consuming work W .

Key Equations Energy Conservation:

$$W + Q_C = Q_H.$$

Coefficient of Performance (COP):

- For a refrigerator (cooling focus):

$$\text{COP}_{\text{refrigerator}} = \frac{Q_C}{W} = \frac{Q_C}{Q_H - Q_C}.$$

- For a heat pump (heating focus):

$$\text{COP}_{\text{heat pump}} = \frac{Q_H}{W} = \frac{Q_H}{Q_H - Q_C}.$$

For an ideal (Carnot) refrigerator or heat pump operating reversibly,

$$\frac{Q_C}{T_C} = \frac{Q_H}{T_H},$$

which gives

$$\text{COP}_{\text{refrigerator, Carnot}} = \frac{T_C}{T_H - T_C}, \quad \text{COP}_{\text{heat pump, Carnot}} = \frac{T_H}{T_H - T_C}.$$

Since $T_H > T_C$, $\text{COP}_{\text{heat pump}} > 1$, making heat pumps more efficient than direct electric heaters (which have $\text{COP} = 1$).

For a refrigerator operating near $T_C \rightarrow 0$, the coefficient of performance (COP) approaches zero, so that an increasingly large amount of work is required to extract even a small amount of heat from the cold reservoir. In the limit $T_C \rightarrow 0$, the work needed becomes unbounded, **making it impossible to reach absolute zero** by any finite process. This directly reflects the unattainability formulation of the **Third Law of Thermodynamics**.

Examples of Refrigerators and Heat Pumps

- Household refrigerators
- Air conditioners
- Heat pumps used for home heating

5.3 Comparison Between a Heat Engine and a Refrigerator

Feature	Heat Engine	Refrigerator
Purpose	Converts heat into work	Transfers heat from cold to hot
Direction of Heat Flow	Hot $T_H \rightarrow$ Cold T_C	Cold $T_C \rightarrow$ Hot T_H
Energy Input	Heat Q_H	Work W
Energy Output	Work W	Heat Q_H
Performance Measure	Efficiency η	Coefficient of Performance (COP)

6 Clausius Theorem

6.1 Clausius Theorem

Theorem 2 (Clausius Theorem (Clausius Inequality)). *For any cyclic process exchanging infinitesimal amounts of heat δQ with heat reservoirs at temperatures T , the cyclic integral satisfies*

$$\oint \frac{\delta Q}{T} \leq 0,$$

where equality holds if and only if the cycle is reversible.

Note 1: The infinitesimal heat exchange δQ is positive ($\delta Q > 0$) when the system absorbs heat from the reservoir, and negative ($\delta Q < 0$) when the system releases heat to the reservoir.

Note 2: It is important to note that a *cyclic process* does not necessarily correspond to a heat engine. A cyclic process is any thermodynamic process in which the system eventually returns to its initial state, so that the net change in internal energy is zero, $\Delta U = 0$. By the First Law of Thermodynamics, the net work done over one cycle is then equal to the total heat exchanged with all reservoirs, $W = \sum_i Q_i$. A heat engine is a special type of cyclic process designed to absorb heat from a high-temperature reservoir, reject some heat to a low-temperature reservoir, and produce net positive work. In contrast, an arbitrary cyclic process may absorb and release heat in any pattern, and may require net work input or output. In the proof of Clausius' theorem, the cyclic process $\mathcal{O}$ is arbitrary and need not be a heat engine; the analysis relies only on its heat exchanges with multiple reservoirs.

Proof. Consider a system undergoing an arbitrary **cyclic** process $\mathcal{O}$, which exchanges heat Q_i with n reservoirs at temperatures $T_1, T_2, \dots, T_n$.

For each reservoir T_i , construct a virtual Carnot engine C_i (also **cyclic**) operating between T_i and a **reference reservoir at temperature** T_0 , where $T_0 \geq T_i$ for all i . The Carnot engine C_i :

- absorbs an amount of heat $Q_i^{(0)}$ from the reference reservoir at T_0 ,
- rejects an amount of heat Q_i to the reservoir at temperature T_i .

By the definition of the absolute temperature scale (from Carnot's theorem), the heat flows satisfy

$$\frac{Q_i^{(0)}}{Q_i} = \frac{T_0}{T_i}.$$

Now operate the cyclic process $\mathcal{O}$ together with the aggregate of auxiliary Carnot engines

$$\Sigma := \sum_{i=1}^n C_i.$$

Each engine C_i is arranged so that its heat exchange with the reservoir at temperature T_i exactly compensates that of the system undergoing $\mathcal{O}$: if the system absorbs an amount Q_i of heat from the reservoir T_i , then C_i supplies the same amount Q_i to that reservoir; if the system releases Q_i to T_i , then C_i absorbs Q_i from it. Consequently, for every reservoir T_i , the net heat exchange due to the combined operation of $\mathcal{O} + \Sigma$ vanishes.

The only remaining thermal interaction of the combined cyclic device is with the reference reservoir at temperature T_0 . The total heat absorbed from this reservoir is therefore

$$Q_0 = \sum_{i=1}^n Q_i^{(0)} = T_0 \sum_{i=1}^n \frac{Q_i}{T_i}.$$

Thus, the entire combined system absorbs heat Q_0 from the single reservoir T_0 and converts it completely into work, producing no other net effect.

According to the Second Law of Thermodynamics (Kelvin-Planck statement), this is impossible unless

$$Q_0 \leq 0,$$

which implies

$$\sum_{i=1}^n \frac{Q_i}{T_i} \leq 0.$$

This proves the inequality part of Clausius theorem.

We now show that the equality sign applies when the cyclic process is reversible. Suppose the cyclic process $\mathcal{O}$ is reversible. Repeat the construction above, but now run the reverse cycle $\mathcal{O}^{-1}$ (the time-reverse of $\mathcal{O}$) together with the corresponding reversed Carnot engines C_i^{-1} . For each i the reversed Carnot engine C_i^{-1} operates between the same pair of reservoirs T_0 and T_i but with reversed heat flows, so that

$$\frac{Q_i^{(0,-)}}{Q_i^-} = \frac{T_0}{T_i},$$

where $Q_i^- = -Q_i$ and $Q_i^{(0,-)} = -Q_i^{(0)}$ are the heats in the reversed construction. Running $\mathcal{O}^{-1}$ together with all $\mathcal{C}_i^{-1}$ cancels the original heat exchanges with each T_i (as before) and yields a net heat

$$Q_0^- = \sum_i Q_i^{(0,-)} = - \sum_i Q_i^{(0)} = -Q_0$$

exchanged with the reference reservoir T_0 .

From the Kelvin–Planck statement the forward combined device $(\mathcal{O} + \Sigma)$ cannot convert a positive heat $Q_0 > 0$ drawn from a single reservoir entirely into work, hence $Q_0 \leq 0$. Likewise, the reversed combined device $(\mathcal{O}^{-1} + \Sigma^{-1})$ cannot convert a positive heat $Q_0^- > 0$ entirely into work, hence $Q_0^- \leq 0$. Together these two inequalities give

$$Q_0 \leq 0 \quad \text{and} \quad -Q_0 = Q_0^- \leq 0,$$

so $Q_0 = 0$. Using $Q_0 = T_0 \sum_i Q_i / T_i$ we obtain

$$\sum_i \frac{Q_i}{T_i} = 0,$$

which proves the equality in Clausius' theorem for a reversible cycle $\mathcal{O}$.

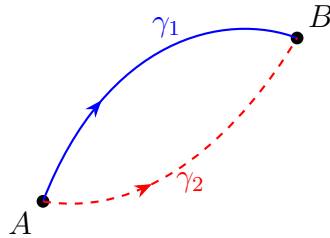
□

6.2 Corollary of Clausius Theorem

Corollary 2. *For any reversible process between two equilibrium states, the integral*

$$\int \frac{\delta Q}{T}$$

is path-independent and depends only on the initial and final states.



Proof. Let A and B be two equilibrium states and let γ_1 and γ_2 be two *reversible* processes connecting A to B . Construct a closed cycle Γ by following γ_1 from A to B , and then following γ_2^- (the reverse of γ_2) from B back to A :

$$\Gamma = \gamma_1 + \gamma_2^-.$$

By the Clausius theorem, for any reversible cycle,

$$\oint_{\Gamma} \frac{\delta Q}{T} = 0.$$

The cyclic integral can be split into contributions from γ_1 and γ_2^- :

$$0 = \oint_{\Gamma} \frac{\delta Q}{T} = \int_{\gamma_1} \frac{\delta Q}{T} + \int_{\gamma_2^-} \frac{\delta Q}{T} = \int_{\gamma_1} \frac{\delta Q}{T} - \int_{\gamma_2} \frac{\delta Q}{T}.$$

Thus,

$$\int_{\gamma_1} \frac{\delta Q}{T} = \int_{\gamma_2} \frac{\delta Q}{T}.$$

Since the integrals along any two reversible paths between the same end states are equal, the value of $\int \frac{\delta Q}{T}$ depends only on the end states, not on the path.

□

7 Entropy

7.1 Concept of Entropy

The corollary of the Clausius theorem established that, for a reversible process between two equilibrium states, the integral

$$\int \frac{\delta Q_{\text{rev}}}{T}$$

is independent of the path taken and depends only on the initial and final states. This means that the quantity

$$\frac{\delta Q_{\text{rev}}}{T}$$

is an *exact differential* of some thermodynamic state function. We call this state function the **entropy**, denoted by S .

Specifically, let A and B be two equilibrium states. The change in entropy between them is defined by

$$S(B) - S(A) := \int_A^B \frac{\delta Q_{\text{rev}}}{T}.$$

Since the integral is path-independent for reversible processes, the value of S is determined uniquely (up to an additive constant) for each thermodynamic state. In differential form, we write

$$dS = \frac{\delta Q_{\text{rev}}}{T}.$$

7.2 Clausius Inequality: Entropy Form

Theorem 3 (Entropy Form of the Clausius Inequality). *For any process (possibly irreversible) taking the system from state A to state B ,*

$$\Delta S := S(B) - S(A) \geq \int_A^B \frac{\delta Q}{T},$$

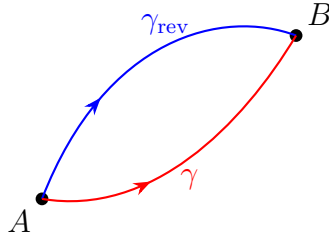
with equality holding if and only if the process is reversible. Here, S is the entropy, a state function defined via the reversible integral.

Proof. Let the system undergo an arbitrary (possibly irreversible) process γ from A to B , exchanging heat with one or more reservoirs. Next, consider any *reversible* process γ_{rev} that also connects A to B . Let γ_{rev}^- denote the same reversible path traversed in the *opposite direction*, i.e., from B to A .

Then the combination

$$\Gamma = \gamma + \gamma_{\text{rev}}^-$$

forms a cyclic process, starting and ending at A .



By the Clausius inequality for cyclic processes (a statement of the second law),

$$\oint_{\Gamma} \frac{\delta Q}{T} \leq 0,$$

with equality if and only if the entire cycle is reversible. Splitting the cycle integral into the two segments gives

$$\int_{\gamma} \frac{\delta Q}{T} + \int_{\gamma_{\text{rev}}^-} \frac{\delta Q_{\text{rev}}}{T} \leq 0.$$

Rewriting the second term and using the definition of entropy for reversible paths,

$$\int_{\gamma_{\text{rev}}^-} \frac{\delta Q_{\text{rev}}}{T} = - \int_{\gamma_{\text{rev}}} \frac{\delta Q_{\text{rev}}}{T} = -(S(B) - S(A)) = -\Delta S,$$

we obtain

$$\int_{\gamma} \frac{\delta Q}{T} - \Delta S \leq 0,$$

or equivalently

$$\Delta S \geq \int_{\gamma} \frac{\delta Q}{T}.$$

If the actual process γ is itself reversible, then the whole cycle Γ is reversible and the Clausius inequality becomes an equality; hence for reversible processes

$$\Delta S = \int_{\gamma} \frac{\delta Q_{\text{rev}}}{T}.$$

Writing the result in an infinitesimal form for a (possibly irreversible) process gives

$$dS \geq \frac{\delta Q}{T}.$$

□

8 On Processes and Entropy of the Universe

8.1 Foundation

Let us recapitulate the fundamental definitions that form the cornerstone of thermodynamic analysis. These concepts are essential for understanding the relationship between process pathways and entropy generation in the universe.

Definition 8.1 (Thermodynamic Process). A **thermodynamic process** describes the evolution of a system from an initial equilibrium state to a final equilibrium state. Depending on how the system traverses **intermediate states**, processes are classified as follows:

- **Reversible Process:** An idealized process in which the system passes through a continuous sequence of equilibrium states. Such a process can be reversed without leaving any net change in either the system or its surroundings. Reversible processes are a conceptual tool for defining maximum work and entropy changes.
- **Irreversible Process:** Any real process in which the system passes through non-equilibrium states. Finite driving forces (e.g., temperature or pressure differences) cause dissipative effects, such as friction, viscous flow, or heat conduction. Irreversible processes generate entropy, and their reversal would produce net changes in the universe.

Definition 8.2 (Thermodynamic Cycle). A **thermodynamic cycle** is a sequence of processes in which a **system** undergoes a series of state transformations and ultimately returns to its initial equilibrium state.

A cycle is considered complete when all intensive and extensive properties of the *system* are restored to their original values. Consequently, for any state function X of the system (such as internal energy U , entropy S , or enthalpy H), the net change over the cycle is zero:

$$\oint dX = 0. \quad (2)$$

The processes composing the cycle may be either *reversible* or *irreversible*.

Remark 1. The condition $\oint dX = 0$ for a state function X is a consequence of the system returning to its initial state, not the definition of a cycle.

Mathematically, the cyclic integral of an exact differential is zero, which is the defining property of a state function. In contrast, path functions like heat Q or work W are not exact differentials, and their cyclic integrals are generally nonzero:

$$\oint dQ \neq 0, \quad \oint dW \neq 0.$$

Nonetheless, the First Law of Thermodynamics, $\oint dU = 0$, implies that over a complete cycle,

$$\oint dQ = \oint dW,$$

showing that the net heat transfer equals the net work done by the system.

Note 1. This definition is **system-centric**: it applies solely to the thermodynamic system. A cycle is said to be complete when all the system's state variables (P, V, T, U, S , etc.) return to their initial values.

Note 2. There is no requirement that the surroundings also return to their original state. In fact, for all real (irreversible) cycles, the surroundings are permanently altered even though the system itself completes a full cycle.

Definition 8.3 (Entropy Change). For any system undergoing a process between equilibrium states 1 and 2, the entropy change is defined as:

$$\Delta S_{\text{system}} = \int_1^2 \frac{\delta Q_{\text{rev}}}{T} \quad (3)$$

where the integral is evaluated along any **reversible** path connecting the two states, δQ_{rev} is the heat transfer to the system, and T is the absolute temperature.

Definition 8.4 (Universe Entropy). The entropy change of the universe is the sum of entropy changes of the system and its surroundings:

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \quad (4)$$

8.2 Main Theorem

Theorem 4 (Non-negativity of Universe Entropy). *For any physical process, the total change of entropy of the universe cannot be negative:*

$$\Delta S_{\text{universe}} \geq 0.$$

That is, it is impossible for any process to decrease the total entropy of the universe.

Proof. Consider a system undergoing an arbitrary process between two equilibrium states. Let δQ denote the infinitesimal heat transferred to the system at a boundary point with temperature T_{boundary} .

Step 1: System entropy change. Construct a reversible process that connects the same initial and final equilibrium states as the actual process. For that reversible path,

$$\Delta S_{\text{system}} = \int \frac{\delta Q_{\text{rev}}}{T},$$

where δQ_{rev} is the heat transferred reversibly. The actual process may be irreversible, but since entropy is a state function, ΔS_{system} has the same value for both the actual and the reversible paths.

Step 2: Clausius inequality. The Clausius inequality states that for any process process,

$$\Delta S_{\text{system}} \geq \int_1^2 \frac{\delta Q}{T_{\text{boundary}}}.$$

This inequality holds for any process, with equality if and only if the process is reversible.

Step 3: Surroundings entropy change. Model the immediate surroundings as an ideal heat reservoir at temperature T_{surr} . The heat transferred to the reservoir is $-\delta Q$, and because the reservoir is assumed to undergo only reversible internal changes,

$$\Delta S_{\text{surroundings}} = \int \frac{-\delta Q}{T_{\text{surr}}}.$$

Step 4: Total universe entropy change. If the heat transfer occurs at a boundary in contact with the reservoir, then $T_{\text{boundary}} = T_{\text{surr}}$. Hence,

$$\begin{aligned} \Delta S_{\text{universe}} &= \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \\ &\geq \int_1^2 \frac{\delta Q}{T_{\text{boundary}}} + \int_1^2 \frac{-\delta Q}{T_{\text{surr}}} \\ &= \int_1^2 \frac{\delta Q}{T_{\text{boundary}}} - \int_1^2 \frac{\delta Q}{T_{\text{boundary}}} = 0. \end{aligned}$$

Thus $\Delta S_{\text{universe}} \geq 0$ for any process, completing the proof. $\square$

Theorem 5 (Entropy Criterion for Thermodynamic Processes). *For a thermodynamic process connecting two equilibrium states of a system, the following equivalences hold:*

(i) *The process is **reversible** if and only if*

$$\Delta S_{\text{universe}} = 0. \tag{5}$$

(ii) *The process is **irreversible** if and only if*

$$\Delta S_{\text{universe}} > 0. \tag{6}$$

Proof. We prove the three logically connected statements:

$$\text{reversible} \iff \Delta S_{\text{universe}} = 0 \quad \text{and} \quad \text{irreversible} \iff \Delta S_{\text{universe}} > 0.$$

If the process is reversible, then $\Delta S_{\text{universe}} = 0$.

Consider a reversible process between equilibrium states 1 and 2.

- *System entropy change:* For a reversible process, the system's entropy change equals the Clausius integral:

$$\Delta S_{\text{system}} = \int_1^2 \frac{\delta Q_{\text{rev}}}{T},$$

where δQ_{rev} is the heat received reversibly by the system and T is the instantaneous system temperature.

- *Surroundings entropy change:* Since the process is reversible, the surroundings remain in thermal equilibrium with the system ($T_{\text{surr}} = T$) and receive heat $\delta Q_{\text{surr}} = -\delta Q_{\text{rev}}$. The surroundings themselves undergo a reversible change, so

$$\Delta S_{\text{surroundings}} = \int_1^2 \frac{\delta Q_{\text{surr}}}{T_{\text{surr}}} = \int_1^2 \frac{-\delta Q_{\text{rev}}}{T}.$$

- *Universe entropy change:* Adding these contributions yields

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} = \int_1^2 \frac{\delta Q_{\text{rev}}}{T} + \int_1^2 \frac{-\delta Q_{\text{rev}}}{T} = 0.$$

Thus reversibility implies $\Delta S_{\text{universe}} = 0$.

If $\Delta S_{\text{universe}} = 0$, then the process is reversible.

We prove the contrapositive: if a process is irreversible, then $\Delta S_{\text{universe}} > 0$.

Consider an irreversible process between the same two equilibrium states. The Clausius inequality (Second Law) gives

$$\Delta S_{\text{system}} > \int_1^2 \frac{\delta Q}{T_{\text{boundary}}},$$

where δQ is the actual heat transferred during the irreversible process and T_{boundary} is the temperature at the system boundary (strict inequality because the process is irreversible).

For the surroundings, if we model them as an *ideal heat reservoir* at constant temperature T_{surr} , then all processes within the reservoir are internally reversible. For such a reservoir,

$$\Delta S_{\text{surroundings}} = \frac{Q_{\text{surr}}}{T_{\text{surr}}} = -\frac{1}{T_{\text{surr}}} \int_1^2 \delta Q.$$

If heat transfer occurs at the boundary in contact with the reservoir, then $T_{\text{boundary}} = T_{\text{surr}}$. Consequently,

$$\begin{aligned}\Delta S_{\text{universe}} &= \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \\ &> \int_1^2 \frac{\delta Q}{T_{\text{boundary}}} - \int_1^2 \frac{\delta Q}{T_{\text{boundary}}} = 0.\end{aligned}$$

Thus irreversibility implies $\Delta S_{\text{universe}} > 0$, which establishes the contrapositive.

Therefore, $\Delta S_{\text{universe}} = 0$ implies the process is reversible.

Equivalence for irreversible processes:

From the above we have shown:

- Reversible $\iff \Delta S_{\text{universe}} = 0$
- Irreversible $\implies \Delta S_{\text{universe}} > 0$

To complete the equivalence, we need: $\Delta S_{\text{universe}} > 0 \implies \text{irreversible}$. This follows logically: if $\Delta S_{\text{universe}} > 0$, the process cannot be reversible (since reversible gives $\Delta S_{\text{universe}} = 0$), so it must be irreversible.

Thus both equivalences (i) and (ii) are proved. $\square$

Remark 2. When we speak of two equilibrium states A and B , we are referring to the system only. The entropy change of the system,

$$\Delta S_{\text{system}}(A \rightarrow B),$$

is a state function: it depends solely on the initial and final equilibrium states and is therefore the same for any process connecting them.

The entropy change of the surroundings, however, is not a state function of the system. It depends on how the process is conducted, because the heat exchanged with the surroundings,

$$\Delta S_{\text{surroundings}} = - \int \frac{\delta Q_{\text{system}}}{T_{\text{surr}}},$$

involves the path-dependent heat transfer δQ_{system} . Consequently, the total entropy change of the universe,

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}},$$

is path-dependent. For the same initial and final system states:

- for a reversible path: $\Delta S_{\text{universe}} = 0$,
- for an irreversible path: $\Delta S_{\text{universe}} > 0$.

The magnitude of $\Delta S_{\text{universe}}$ quantifies the irreversibility of the particular path taken.

9 The Role of Process Information in Determining Entropy Change

The calculation of entropy change ΔS in thermodynamic systems often causes confusion regarding its dependence on the process path. This section clarifies the nuanced relationship between entropy as a state function and the role of process information in determining thermodynamic states.

9.1 Entropy as a State Function

The fundamental definition of entropy change for a reversible process is:

$$\Delta S = \int \frac{\delta Q_{\text{rev}}}{T}. \quad (7)$$

Crucially, **entropy S is a state function**, meaning:

$$\Delta S = S_B - S_A \quad (8)$$

depends **only** on the initial state A and final state B, not on the path taken between them. This is the mathematical expression of entropy being a state function.

9.2 The Role of Process Information

The definition of entropy originates from the Clausius theorem, which applies to reversible cycles,

$$\oint_{\text{rev}} \frac{\delta Q_{\text{rev}}}{T} = 0.$$

This relation implies the existence of a state function S such that, for any reversible process between two equilibrium states,

$$\Delta S = S_2 - S_1 = \int_1^2 \frac{\delta Q_{\text{rev}}}{T}.$$

Although the entropy change ΔS is *path-independent*, knowledge of the specific *process* remains essential to determine the *final state* of the system. Different thermodynamic processes, even when starting from the same initial conditions, generally lead to distinct final states. In practice, we are often interested in questions such as: “What is the entropy change associated with a given process?”—which implicitly defines the final state of the system. Once both states are well-defined equilibrium states of the same system and can be connected by at least one reversible path, the entropy change can be evaluated by any convenient reversible route. However, if no reversible path exists between the

two equilibrium states—for instance, if they belong to different phases or correspond to different systems—then ΔS is not defined within classical thermodynamics.

For example:

- **Free expansion:** $Q = 0, W = 0 \Rightarrow \Delta U = 0 \Rightarrow T = \text{constant}$
- **Reversible adiabatic expansion:** $Q = 0 \Rightarrow \Delta S = 0$
- **Isothermal process:** $T = \text{constant}$ by definition

Hence, for an ideal gas initially at (P_1, V_1, T_1) , the corresponding final states can be summarized as:

$$\text{Free expansion: } (P_2, V_2, T_1) \quad (9)$$

$$\text{Adiabatic expansion: } (P_2, V_2, T_2), \quad \text{with } T_2 < T_1 \quad (10)$$

$$\text{Isothermal compression: } (P_2, V_2, T_1) \quad (11)$$

Thus, even though ΔS depends only on the initial and final states, the process information determines what that final state actually is.

9.3 Principle of Thermodynamic Connectability

Two equilibrium states of a closed system are said to be **thermodynamically connectable** if one can be transformed into the other through some physically realizable **reversible** process. Such connectability requires that the following necessary fundamental principles be satisfied:

1. **Identical System Composition** The system must have the same chemical composition and total mass in both states. A state with one mole of gas cannot be connected to a state with two moles of gas by any thermodynamic process.
2. **Consistency with a Single Equation of State** Both states must represent valid, stable equilibrium points described by the same fundamental relation or equation of state.

Example 1. For an ideal gas, both states must satisfy

$$PV = nRT.$$

One cannot connect a state that obeys the ideal gas law to a state that follows, for instance, the van der Waals equation.

3. **Obedience to the First Law (Energy Conservation)** The transformation between the two states must satisfy the First Law of Thermodynamics,

$$\Delta U = Q - W,$$

where ΔU is the change in internal energy, Q is the heat exchanged with the surroundings, and W is the work done by the system. Any admissible process must be consistent with this energy balance.

4. **Consistency with the Second Law** The reversible transformation between the two states must satisfy the Second Law of Thermodynamics. For a reversible process, the total entropy change of the system and surroundings is zero:

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} = 0.$$

States that cannot be connected by a reversible process without violating this condition are considered **thermodynamically unconnectable**.

10 Examples of Entropy Change

When calculating the entropy change between two equilibrium states, the actual process taken by the system is irrelevant. Entropy is a **state function**, and its change depends only on the initial and final states. By definition,

$$\Delta S = \int_A^B \frac{\delta Q_{\text{rev}}}{T},$$

where the integration is carried out over any *reversible* path connecting the same two states. The real process, whether reversible or irreversible, does not appear in this expression. One may therefore choose a convenient reversible path—even if it is purely hypothetical—to compute ΔS .

10.1 Reversible Isothermal Expansion of an Ideal Gas

Consider an ideal gas containing n moles undergoing a reversible isothermal expansion from volume V_1 to V_2 at temperature T . Since the process is isothermal, the internal energy U remains constant:

$$\Delta U = 0.$$

From the first law of thermodynamics,

$$\Delta U = Q - W = 0 \implies Q = W.$$

For a reversible isothermal expansion, the work done by the gas is

$$W = \int_{V_1}^{V_2} P dV.$$

Using the ideal gas law $P = \frac{nRT}{V}$, we get

$$W = \int_{V_1}^{V_2} \frac{nRT}{V} dV = nRT \ln \frac{V_2}{V_1}.$$

Since the process is reversible, the entropy change of the gas is

$$\Delta S = \int \frac{\delta Q_{\text{rev}}}{T} = \frac{Q}{T} = \frac{W}{T} = nR \ln \frac{V_2}{V_1}.$$

This means the entropy increases if the volume expands ($V_2 > V_1$).

Since this process is reversible, the entropy change of the surroundings exactly cancels that of the system, yielding

$$\Delta S_{\text{universe}} = 0,$$

as required.

10.2 Free Expansion of an Ideal Gas (Irreversible Process)

In free expansion, an ideal gas expands suddenly into a vacuum without performing work and without exchanging heat with the surroundings:

$$Q = 0, \quad W = 0.$$

Since the system is isolated and no work is done, the internal energy remains constant:

$$\Delta U = Q - W = 0.$$

For an ideal gas, internal energy depends only on temperature (see Appendix A), so **the temperature remains constant during free expansion**.

Despite the absence of heat exchange, the entropy of the gas increases. Since the process is irreversible, the entropy change ΔS_{sys} cannot be calculated directly from the Clausius integral along a real path γ ,

$$\Delta S_{\text{sys}} \neq \int_{\gamma} \frac{\delta Q}{T}.$$

To evaluate ΔS_{sys} , we invoke a **hypothetical reversible path** connecting the same initial and final equilibrium states. Because the temperature remains constant, the most convenient choice is a reversible isothermal expansion from volume V_1 to V_2 . Along this reversible path, the entropy change of the system is

$$\Delta S_{\text{sys}} = nR \ln \frac{V_2}{V_1} > 0.$$

This result depends only on the end states and therefore also applies to the actual free expansion, even though that process itself is irreversible.

In a free expansion, the gas expands into a vacuum and exchanges no heat with the surroundings:

$$Q = 0 \quad \Rightarrow \quad \Delta S_{\text{surr}} = 0.$$

Consequently, the total entropy change of the universe is

$$\Delta S_{\text{universe}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = nR \ln \frac{V_2}{V_1} > 0.$$

Unlike a reversible isothermal expansion, for which the entropy change of the universe vanishes, the free expansion leads to a positive entropy production. This increase in the entropy of the universe is the hallmark of irreversibility and provides a clear thermodynamic signature of spontaneous free expansion.

10.3 Heat Conduction Between Two Reservoirs

Consider two reservoirs: a hot reservoir at temperature T_H and a cold reservoir at temperature T_C ($T_H > T_C$). A certain amount of heat Q is transferred from the hot to the cold reservoir. We compare two cases.

(a) Natural conduction (irreversible process): Heat Q flows spontaneously through a conducting rod. The system (the rod) simply conducts heat and, after steady state is reached, returns to its original state at the end of the process, so

$$\Delta S_{\text{system}} = 0.$$

The entropy changes of the reservoirs are

$$\Delta S_H = -\frac{Q}{T_H}, \quad \Delta S_C = +\frac{Q}{T_C}.$$

The total (universe) entropy change is therefore

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_H + \Delta S_C = Q \left(\frac{1}{T_C} - \frac{1}{T_H} \right) > 0.$$

Hence, natural conduction is **irreversible** and produces entropy.

(b) Reversible transfer via a Carnot engine: Now insert a reversible (Carnot) engine between the same two reservoirs. During one complete cycle, the engine absorbs heat Q_H from the hot reservoir, produces work W , and rejects heat Q_C to the cold reservoir:

$$Q_H = Q_C + W.$$

- **Entropy change of the engine (system):** Since the engine completes a full cycle, it returns to its original thermodynamic state:

$$\Delta S_{\text{engine}} = 0.$$

- **Entropy changes of the reservoirs:** The hot reservoir loses heat Q_H and the cold reservoir gains heat Q_C :

$$\Delta S_H = -\frac{Q_H}{T_H}, \quad \Delta S_C = \frac{Q_C}{T_C}.$$

For a reversible process, the heat flows satisfy

$$\frac{Q_H}{T_H} = \frac{Q_C}{T_C}.$$

- **Total entropy change of the universe:** The sum of all entropy changes is

$$\Delta S_{\text{universe}} = \Delta S_{\text{engine}} + \Delta S_H + \Delta S_C = 0 + \left(-\frac{Q_H}{T_H}\right) + \frac{Q_C}{T_C} = 0.$$

No entropy is produced, indicating a fully reversible process.

Conclusion: In both cases, the system (rod or engine) returns to its initial state, so only the reservoirs contribute to the net entropy change. The difference lies in the reversibility of the process:

$$\begin{cases} \text{Natural conduction:} & \Delta S_{\text{universe}} > 0, \\ \text{Carnot engine:} & \Delta S_{\text{universe}} = 0, \end{cases}$$

as expected.

10.4 Entropy of an Ideal Gas

The entropy S of an ideal gas can be expressed as a function of temperature T and volume V . Starting from the first law for a reversible process,

$$\delta Q_{\text{rev}} = dU + P dV.$$

For an ideal gas, the internal energy is a state function that depends only on temperature. Since dU is path independent, it may be evaluated along any convenient reversible path connecting neighboring equilibrium states. In particular, choosing a constant-volume path ($dV = 0$) gives $\delta Q = dU$ by the first law, and by definition of the molar heat capacity at constant volume, $\delta Q = nc_V dT$. Therefore,

$$dU = nc_V dT,$$

where c_V is the molar heat capacity at constant volume and n is the mole number. Thus

$$\delta Q_{\text{rev}} = nc_V dT + P dV.$$

Substituting the ideal gas law $P = \frac{nRT}{V}$,

$$\delta Q_{\text{rev}} = nc_V dT + \frac{nRT}{V} dV.$$

Dividing by T , the differential change in entropy is

$$dS = \frac{\delta Q_{\text{rev}}}{T} = nc_V \frac{dT}{T} + nR \frac{dV}{V}. \quad (12)$$

Integrating from a reference state (T_0, V_0) to (T, V) ,

$$S(T, V) = S_0 + nc_V \ln \frac{T}{T_0} + nR \ln \frac{V}{V_0},$$

where S_0 is the entropy at the reference state.

Alternatively, starting from the ideal gas law,

$$P = \frac{nRT}{V},$$

we can express the volume as

$$V = \frac{nRT}{P}.$$

Taking the differential, we have

$$dV = \frac{nR}{P}dT - \frac{nRT}{P^2}dP.$$

Using this, the entropy differential dS in Eq. (12) can be rewritten as a function of T and P :

$$dS = nc_P \frac{dT}{T} - nR \frac{dP}{P},$$

where $c_P = c_V + R$ is the molar heat capacity at constant pressure, and R is the gas constant.

Integrating from a reference state (T_0, P_0) , we get

$$S(T, P) = S_0 + nc_P \ln \frac{T}{T_0} - nR \ln \frac{P}{P_0}.$$

These expressions illustrate the fundamental role of entropy in quantifying energy dispersal and irreversibility in thermodynamic processes.

11 Thermodynamic Potentials

11.1 Definitions and Basic Relations

Thermodynamic potentials characterize the equilibrium properties of a system subject to specified external constraints. Each potential is most naturally expressed in terms of a particular set of independent variables, called its *natural variables*, which correspond to the thermodynamic quantities controlled or held fixed by the environment in the class of physical processes for which that potential is especially convenient. Once a thermodynamic potential is known as a function of its natural variables, all other thermodynamic properties of the system follow directly from appropriate partial derivatives.

The internal energy U is a fundamental thermodynamic potential, naturally expressed as a function of entropy S , volume V , and particle number N :

$$U = U(S, V, N).$$

Combining the first and second laws, the total differential of U is¹.

$$dU = T dS - P dV + \mu dN,$$

which explicitly shows that (S, V, N) are the natural variables of U . The coefficients of these differentials correspond to the conjugate intensive quantities: temperature T , pressure P , and chemical potential μ , where μ is defined as

$$\mu := \left(\frac{\partial U}{\partial N} \right)_{S,V}.$$

The chemical potential μ can be viewed as a potential for energy gain, analogous to how temperature T governs heat flow and pressure P governs mechanical work. It represents the tendency of a system to exchange particles: in equilibrium, particles flow from regions of higher to lower chemical potential, just as heat flows from higher to lower temperature. Thus, μ serves as the driving “potential” associated with changes in particle number and the corresponding energy transfer.

Table 1 presents the three fundamental modes of energy transfer into the system: thermal, mechanical, and chemical.

Table 1: Energy Transfer into the System

Transfer type	Symbol	Description
Heat	δQ	Energy transfer due to temperature difference
Mechanical work	$-P dV$	Energy transfer from volume compression or expansion
Chemical work	μdN	Energy carried in by incoming particles (chemical work)

Taking partial derivatives, we obtain

$$\left(\frac{\partial U}{\partial S} \right)_{V,N} = T, \quad \left(\frac{\partial U}{\partial V} \right)_{S,N} = -P, \quad \left(\frac{\partial U}{\partial N} \right)_{S,V} = \mu.$$

These express that temperature, pressure, and chemical potential can be obtained from the internal energy.

Building upon the internal energy U , we can define various important thermodynamic potentials that are better suited to describe systems under different experimental conditions. The mathematical framework behind these definitions is the theory of *Legendre transformations*, which allows us to systematically change the set of independent variables of a thermodynamic function. This transformation introduces new potentials whose *natural variables* correspond directly to physically controllable parameters, such as temperature or pressure.

Legendre transformations provide a rigorous and elegant way to shift from one thermodynamic description to another while preserving all the thermodynamic information. We

¹For a system that can exchange particles, the First Law reads

$$dU = \delta Q - \delta W + (\text{energy carried by particles}),$$

where the last term accounts for the energy brought into the system by particle transfer.

will discuss the theory and application of Legendre transformations in detail separately, as it forms the foundation for understanding the natural variables and the relationships between different thermodynamic potentials.

All thermodynamic potentials are, in principle, equivalent descriptions of the same underlying system. Choosing a particular potential is analogous to selecting a representation in quantum mechanics: each offers a convenient framework tailored to specific experimental constraints or variables held fixed. The differences lie not in the physics but in the choice of natural variables, allowing us to analyze the system most effectively under given conditions.

- **Helmholtz Free Energy F :**

Defined as

$$F(T, V, N) := U(S, V, N) - TS,$$

the **Helmholtz free energy** is obtained from the internal energy U through a **Legendre transformation** that replaces entropy S with temperature T as an independent variable, since

$$\left(\frac{\partial U}{\partial S}\right)_{V,N} = T.$$

Thus, the *natural variables* of F are T , V , and N , making it particularly suitable for describing systems held at constant temperature and volume. Taking the differential of F gives

$$dF = dU - d(TS) = T dS - P dV + \mu dN - T dS - S dT = -S dT - P dV + \mu dN,$$

which confirms that F naturally depends on (T, V, N) . Taking partial derivatives, we obtain

$$\left(\frac{\partial F}{\partial T}\right)_{V,N} = -S, \quad \left(\frac{\partial F}{\partial V}\right)_{T,N} = -P, \quad \left(\frac{\partial F}{\partial N}\right)_{T,V} = \mu.$$

These relations show that the entropy, pressure, and chemical potential can be directly derived from the Helmholtz free energy.

- **Enthalpy H :**

Defined as

$$H(S, P, N) := U(S, V, N) + PV,$$

the **enthalpy** H is obtained from the internal energy U by a **Legendre transformation** that replaces volume V with pressure P as an independent variable, since

$$\left(\frac{\partial U}{\partial V}\right)_{S,N} = -P.$$

Thus, the natural variables of H are (S, P, N) . It is particularly useful for describing processes at constant pressure. Its differential is

$$dH = T dS + V dP + \mu dN,$$

which confirms that H naturally depends on (S, P, N) . Taking partial derivatives, we obtain

$$\left(\frac{\partial H}{\partial S}\right)_{P,N} = T, \quad \left(\frac{\partial H}{\partial P}\right)_{S,N} = V, \quad \left(\frac{\partial H}{\partial N}\right)_{S,P} = \mu.$$

- **Gibbs Free Energy G :**

Defined as

$$G(T, P, N) := U(S, V, N) - TS + PV = H - TS,$$

the **Gibbs free energy** G is obtained from the internal energy U through a **Legendre transformation** that replaces entropy S and volume V with temperature T and pressure P as independent variables. This transformation follows from the relations

$$\left(\frac{\partial U}{\partial S}\right)_{V,N} = T, \quad \left(\frac{\partial U}{\partial V}\right)_{S,N} = -P.$$

The natural variables of G are T , P , and N , which makes it ideal for systems at constant temperature and pressure. Its differential is

$$dG = -S dT + V dP + \mu dN,$$

which confirms that G naturally depends on (T, P, N) . Taking partial derivatives, we obtain

$$\left(\frac{\partial G}{\partial T}\right)_{P,N} = -S, \quad \left(\frac{\partial G}{\partial P}\right)_{T,N} = V, \quad \left(\frac{\partial G}{\partial N}\right)_{T,P} = \mu.$$

- **Grand Potential (or Landau Potential) Φ_G :**

Defined as

$$\Phi_G(T, V, \mu) := U(S, V, N) - TS - \mu N = F - \mu N,$$

the **grand potential** Φ_G is obtained from the internal energy U by a **Legendre transformation** that replaces entropy S and particle number N with temperature T and chemical potential μ as independent variables. This transformation follows from the relations

$$\left(\frac{\partial U}{\partial S}\right)_{V,N} = T, \quad \left(\frac{\partial U}{\partial N}\right)_{S,V} = \mu.$$

The natural variables are temperature T , volume V , and chemical potential μ , appropriate for systems in contact with a particle reservoir at fixed temperature and volume. Its differential is

$$d\Phi_G = -S dT - P dV - N d\mu.$$

which confirms that Φ_G naturally depends on (T, V, μ) . Taking partial derivatives, we obtain

$$\left(\frac{\partial \Phi_G}{\partial T}\right)_{V,\mu} = -S, \quad \left(\frac{\partial \Phi_G}{\partial V}\right)_{T,\mu} = -P, \quad \left(\frac{\partial \Phi_G}{\partial \mu}\right)_{T,V} = -N.$$

Table 2 provides an overview of the thermodynamic potentials, including their definitions and the expressions for their differentials in terms of their natural variables.

Table 2: Thermodynamic potentials.

Potential	Definition	Differential)
Internal Energy U	$U(S, V, N)$	$dU = T dS - P dV + \mu dN$
Helmholtz Free Energy F	$F(T, V, N) = U - TS$	$dF = -S dT - P dV + \mu dN$
Enthalpy H	$H(S, P, N) = U + PV$	$dH = T dS + V dP + \mu dN$
Gibbs Free Energy G	$G(T, P, N) = U - TS + PV$	$dG = -S dT + V dP + \mu dN$
Grand Potential Φ_G	$\Phi_G(T, V, \mu) = U - TS - \mu N$	$d\Phi_G = -S dT - P dV - N d\mu$

11.2 Heat Capacities and Material Response: Basic Definitions

For a simple compressible system (single component, single phase), we denote the thermodynamic variables T (temperature), P (pressure), V (volume), S (entropy), U (internal energy), and $H = U + PV$ (enthalpy). We define

$$C_V \equiv \left(\frac{\partial Q}{\partial T} \right)_V = \left(\frac{\partial U}{\partial T} \right)_V = T \left(\frac{\partial S}{\partial T} \right)_V, \quad (13)$$

$$C_P \equiv \left(\frac{\partial Q}{\partial T} \right)_P = \left(\frac{\partial H}{\partial T} \right)_P = T \left(\frac{\partial S}{\partial T} \right)_P. \quad (14)$$

Here we have used $\delta Q_{\text{rev}} = T dS$ (for a reversible process) together with the first law of thermodynamics $T dS = dU + P dV$. When the particle number N is relevant, the above definitions are understood to be taken **at fixed** N as well.

We define the following experimentally measurable response quantities for a simple compressible system:

$$\alpha \equiv \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P \quad (\text{coefficient of thermal expansion}), \quad (15)$$

$$\kappa_T \equiv -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T \quad (\text{isothermal compressibility}), \quad (16)$$

$$\kappa_S \equiv -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_S \quad (\text{adiabatic compressibility}). \quad (17)$$

For mechanically stable equilibrium states, these coefficients are positive. Specifically, a small increase in temperature at constant pressure generally causes the volume to expand, i.e., $\alpha > 0$.

The isothermal compressibility κ_T characterizes the volume change when the system is compressed slowly enough to maintain a constant temperature by exchanging heat with the surroundings. Under such conditions, the volume decreases as pressure increases, while the temperature remains fixed due to heat release or absorption, ensuring that $\kappa_T > 0$.

The adiabatic compressibility κ_S characterizes volume changes during a *quasi-static adiabatic* compression or expansion, where the process is slow enough to maintain thermo-

dynamic equilibrium but occurs without heat exchange with the surroundings, thereby keeping the entropy constant. Because compression increases the particles' kinetic energy—raising both the temperature and the internal pressure—the system exhibits greater resistance to volume change under these conditions. Consequently, the adiabatic compressibility is generally smaller than the isothermal compressibility:

$$\kappa_S < \kappa_T.$$

This difference is critical in phenomena such as sound propagation, where pressure disturbances travel too fast for heat exchange to occur, making the process effectively adiabatic. The speed of sound c in a medium is related to the adiabatic compressibility by

$$c = \frac{1}{\sqrt{\rho \kappa_S}},$$

where ρ is the mass density; thus, a smaller κ_S corresponds to a stiffer medium and faster sound speed.

11.3 Key Maxwell Relations and Thermodynamic Identities

In thermodynamics, a *Maxwell relation* is an identity between partial derivatives of state variables that arises from the property of exact differentials. If a thermodynamic potential is expressed as a function of its natural variables, it is a *state function*, and therefore its total differential is *exact*. By Schwarz's theorem, the mixed second derivatives of an exact differential are equal, which gives rise to symmetry relations connecting measurable quantities in different experimental contexts.

For the Helmholtz free energy $F = F(T, V, N)$, the total differential is

$$dF = -S dT - P dV + \mu dN,$$

so that

$$S = -\left(\frac{\partial F}{\partial T}\right)_{V,N}, \quad P = -\left(\frac{\partial F}{\partial V}\right)_{T,N}, \quad \mu = \left(\frac{\partial F}{\partial N}\right)_{T,V}.$$

The exactness of dF then implies the following Maxwell relations:

$$\left(\frac{\partial S}{\partial V}\right)_{T,N} = \left(\frac{\partial P}{\partial T}\right)_{V,N}, \quad \left(\frac{\partial S}{\partial N}\right)_{T,V} = -\left(\frac{\partial \mu}{\partial T}\right)_{V,N}, \quad \left(\frac{\partial \mu}{\partial V}\right)_{T,N} = -\left(\frac{\partial P}{\partial N}\right)_{T,V}.$$

Similarly, for the Gibbs free energy $G = G(T, P, N)$, we have

$$dG = -S dT + V dP + \mu dN,$$

leading to

$$S = -\left(\frac{\partial G}{\partial T}\right)_{P,N}, \quad V = \left(\frac{\partial G}{\partial P}\right)_{T,N}, \quad \mu = \left(\frac{\partial G}{\partial N}\right)_{T,P}.$$

From the exactness of dG , the corresponding Maxwell relations follow:

$$\left(\frac{\partial S}{\partial P}\right)_{T,N} = -\left(\frac{\partial V}{\partial T}\right)_{P,N}, \quad \left(\frac{\partial S}{\partial N}\right)_{T,P} = -\left(\frac{\partial \mu}{\partial T}\right)_{P,N}, \quad \left(\frac{\partial V}{\partial N}\right)_{T,P} = \left(\frac{\partial \mu}{\partial P}\right)_{T,N}.$$

These relations are direct consequences of the fact that the differential of a thermodynamic potential is exact, and they provide powerful connections between experimentally accessible quantities.

The implicit function relation for variables P, V, T (with fixed N) gives:

$$\left(\frac{\partial P}{\partial T}\right)_{V,N} = -\frac{\left(\frac{\partial V}{\partial T}\right)_{P,N}}{\left(\frac{\partial V}{\partial P}\right)_{T,N}},$$

a result presented in the lecture notes on the first law of thermodynamics.

Recall that the thermal expansion coefficient α and isothermal compressibility κ_T are:

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_{P,N}, \quad \kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial P}\right)_{T,N}.$$

Hence,

$$\left(\frac{\partial P}{\partial T}\right)_{V,N} = \frac{\alpha}{\kappa_T}.$$

The entropy differential can be expressed as

$$dS = \left(\frac{\partial S}{\partial T}\right)_{V,N} dT + \left(\frac{\partial S}{\partial V}\right)_{T,N} dV + \left(\frac{\partial S}{\partial N}\right)_{T,V} dN,$$

or

$$dS = \left(\frac{\partial S}{\partial T}\right)_{P,N} dT + \left(\frac{\partial S}{\partial P}\right)_{T,N} dP + \left(\frac{\partial S}{\partial N}\right)_{T,P} dN.$$

Using

$$\left(\frac{\partial S}{\partial T}\right)_{V,N} = \frac{C_V}{T}, \quad \left(\frac{\partial S}{\partial T}\right)_{P,N} = \frac{C_P}{T},$$

and the Maxwell relations,

$$\left(\frac{\partial S}{\partial V}\right)_{T,N} = \left(\frac{\partial P}{\partial T}\right)_{V,N}, \quad \left(\frac{\partial S}{\partial P}\right)_{T,N} = -\left(\frac{\partial V}{\partial T}\right)_{P,N},$$

we write:

$$dS(T, V, N) = \frac{C_V}{T} dT + \left(\frac{\partial P}{\partial T}\right)_{V,N} dV + \left(\frac{\partial S}{\partial N}\right)_{T,V} dN, \quad (18)$$

$$dS(T, P, N) = \frac{C_P}{T} dT - \left(\frac{\partial V}{\partial T}\right)_{P,N} dP + \left(\frac{\partial S}{\partial N}\right)_{T,P} dN. \quad (19)$$

Starting from the definition

$$C_P = T \left(\frac{\partial S}{\partial T} \right)_{P,N},$$

we recognize that the dependence of the entropy S on (T, P, N) can be expressed through (T, V, N) :

$$S = S(T, V, N) = S(T, V(T, P, N), N),$$

where the dependence $V = V(T, P, N)$ arises from the fundamental thermodynamic relation for the Gibbs free energy,

$$V = \left(\frac{\partial G}{\partial P} \right)_{T,N}.$$

It follows that

$$\left(\frac{\partial S}{\partial T} \right)_{P,N} = \left(\frac{\partial S}{\partial T} \right)_{V,N} + \left(\frac{\partial S}{\partial V} \right)_{T,N} \left(\frac{\partial V}{\partial T} \right)_{P,N}.$$

Recognizing that

$$\left(\frac{\partial S}{\partial T} \right)_{V,N} = \frac{C_V}{T} \quad \text{and} \quad \left(\frac{\partial S}{\partial V} \right)_{T,N} = \left(\frac{\partial P}{\partial T} \right)_{V,N}$$

(from a Maxwell relation), we obtain

$$\left(\frac{\partial S}{\partial T} \right)_{P,N} = \frac{C_V}{T} + \left(\frac{\partial P}{\partial T} \right)_{V,N} \left(\frac{\partial V}{\partial T} \right)_{P,N}.$$

Multiplying by T , we get:

$$C_P = C_V + T \left(\frac{\partial P}{\partial T} \right)_{V,N} \left(\frac{\partial V}{\partial T} \right)_{P,N}.$$

Using $\left(\frac{\partial V}{\partial T} \right)_{P,N} = V\alpha$ and $\left(\frac{\partial P}{\partial T} \right)_{V,N} = \frac{\alpha}{\kappa_T}$, this yields:

$$\boxed{C_P - C_V = TV \frac{\alpha^2}{\kappa_T}} \tag{20}$$

To analyze the compressibilities, we begin with the differential of the entropy expressed in the variables (T, P, N) , Eq. (19):

$$dS = \frac{C_P}{T} dT - V\alpha dP + \left(\frac{\partial S}{\partial N} \right)_{T,P} dN.$$

Under adiabatic conditions ($dS = 0$) and fixed N , we have:

$$0 = \frac{C_P}{T} dT - V\alpha dP, \quad \Rightarrow \quad \left(\frac{\partial T}{\partial P} \right)_{S,N} = \frac{TV\alpha}{C_P}. \tag{21}$$

The adiabatic compressibility κ_S is:

$$\kappa_S = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_{S,N}.$$

For V as a function of S, P, N via

$$V = V(P, T(S, P, N), N),$$

using the chain rule:

$$\left(\frac{\partial V}{\partial P}\right)_{S,N} = \left(\frac{\partial V}{\partial P}\right)_{T,N} + \left(\frac{\partial V}{\partial T}\right)_{P,N} \left(\frac{\partial T}{\partial P}\right)_{S,N},$$

substitute:

$$\left(\frac{\partial V}{\partial P}\right)_{T,N} = -V\kappa_T, \quad \left(\frac{\partial V}{\partial T}\right)_{P,N} = V\alpha, \quad \left(\frac{\partial T}{\partial P}\right)_{S,N} = \frac{TV\alpha}{C_P},$$

to get:

$$\left(\frac{\partial V}{\partial P}\right)_{S,N} = -V\kappa_T + V\alpha \cdot \frac{TV\alpha}{C_P} = -V\kappa_T + \frac{TV^2\alpha^2}{C_P}.$$

Dividing by $-V$:

$$\kappa_S = \kappa_T - \frac{TV\alpha^2}{C_P},$$

or

$$\boxed{\kappa_T - \kappa_S = \frac{TV\alpha^2}{C_P}}$$

Using Eq. (20) for $C_P - C_V$, rewrite $TV\alpha^2$ as:

$$TV\alpha^2 = (C_P - C_V)\kappa_T,$$

and substitute back:

$$\kappa_S = \kappa_T - \frac{(C_P - C_V)\kappa_T}{C_P} = \kappa_T \left(1 - \frac{C_P - C_V}{C_P}\right) = \kappa_T \frac{C_V}{C_P}.$$

Hence,

$$\boxed{\frac{\kappa_T}{\kappa_S} = \frac{C_P}{C_V}}$$

Alternatively, we can express C_V and C_P in terms of κ_T and κ_S :

$$C_V = \frac{TV\alpha^2\kappa_S}{(\kappa_T - \kappa_S)\kappa_T}, \quad C_P = \frac{TV\alpha^2}{\kappa_T - \kappa_S}.$$

Remark 3.

- For most substances, $C_P > C_V$, thus $\kappa_S < \kappa_T$, meaning systems are less compressible adiabatically than isothermally.
- The ratio $\gamma := C_P/C_V$ is crucial in acoustics and thermodynamics; e.g., the speed of sound c relates to κ_S via $c^2 = 1/(\rho\kappa_S)$ where ρ is the mass density. Knowing κ_T and γ determines $\kappa_S = \kappa_T/\gamma$. Typically, the adiabatic compressibility κ_S is more difficult to measure directly because it corresponds to processes occurring without heat exchange, requiring rapid or carefully controlled adiabatic conditions. In contrast, the isothermal compressibility κ_T is easier to determine experimentally since it involves slow, constant-temperature processes.

- All the above derivations assume fixed particle number N but explicitly including N extends the framework to open systems. Even when the particle number N is fixed, the chemical potential μ remains a well-defined thermodynamic quantity. It is defined as

$$\mu := \left(\frac{\partial U}{\partial N} \right)_{S,V},$$

representing how the internal energy would change if a particle were added while keeping entropy and volume constant. In processes with $dN = 0$, the term μdN drops out of the differential relation $dU = T dS - P dV + \mu dN$, so no energy is exchanged through particle transfer. Nonetheless, μ continues to characterize the system's equilibrium properties, just as pressure P remains meaningful even when the volume does not change.

- Maxwell relations including the chemical potential μ provide insight into chemical equilibrium and phase transitions.

12 Extremum Principles and Spontaneous Processes

12.1 The Fundamental Thermodynamic Inequality

From the First Law of Thermodynamics, which expresses energy conservation, the change in internal energy U is given by

$$dU = \delta Q - P dV + \mu dN.$$

The Second Law of Thermodynamics states that for any infinitesimal process is

$$T dS \geq \delta Q.$$

Combining the two laws,

$$T dS \geq \delta Q = dU + P dV - \mu dN,$$

which rearranges to

$$\boxed{dU \leq T dS - P dV + \mu dN} \quad (22)$$

This fundamental inequality holds for all thermodynamic processes, with equality for reversible ones.

12.2 Extremum Principles for Spontaneous Processes

A *spontaneous process* is one that occurs naturally, without external intervention, once the external constraints are fixed. Depending on which quantities are held constant—such

as energy, temperature, pressure, or volume—the system may exchange heat or work with its surroundings, but it always evolves toward an equilibrium state in accordance with the Second Law of Thermodynamics. More generally, a spontaneous process represents the natural relaxation of a system from a nonequilibrium configuration toward equilibrium. It should be noted that thermodynamic functions are usually defined only for equilibrium states; however, they can be extended to describe nonequilibrium situations, as will be discussed later in Subsection 12.4.

When a system is already in an equilibrium state under given external constraints, it will not undergo any further spontaneous evolution into another equilibrium state. In such a state, all thermodynamic driving forces—such as gradients of temperature, pressure, or chemical potential—have vanished. For example, if the temperature T is uniform, there is no heat flow; if the pressure P is uniform, there is no mechanical work causing expansion or compression; and if the chemical potentials of different components are equal, there is no net diffusion or chemical reaction progress. The appropriate thermodynamic potential has thus reached its extremum value: for instance, the entropy S is maximized in an isolated system, the Helmholtz free energy F is minimized at constant T, V, N , and the Gibbs free energy G is minimized at constant T, P, N . Consequently, $dS = 0$, $dF = 0$, or $dG = 0$, respectively, and no spontaneous change can occur unless the external constraints themselves are altered.

1. Entropy S The Second Law of Thermodynamics states that for any infinitesimal process,

$$T dS \geq \delta Q.$$

In an adiabatic process ($\delta Q = 0$), this reduces to $T dS \geq 0$, which for $T > 0$ implies $dS \geq 0$. Integrating, we obtain

$$\boxed{S \leq S_{\text{eq}}}$$

where S_{eq} denotes the equilibrium entropy. Thus, in an isolated adiabatic system, the entropy can only increase for irreversible processes or remain constant for reversible ones, establishing the thermodynamic arrow of time. In other words, therefore, all equilibrium states that can be connected by reversible adiabatic transformations lie on the same constant-entropy (isoentropic) surface in the thermodynamic state space.

2. Internal Energy $U(S, V, N)$ At fixed entropy S , volume V , and particle number N ,

$$dS = dV = dN = 0,$$

thus from the fundamental inequality Eq. (22),

$$dU \leq 0.$$

Interpretation: Spontaneous changes at fixed (S, V, N) cause the internal energy to decrease or remain constant. Equilibrium corresponds to the minimum internal energy:

$$\boxed{U \geq U_{\text{eq}}}$$

3. Helmholtz Free Energy $F(T, V, N) = U - TS$ Differentiating,

$$dF = dU - T dS - S dT.$$

Using the inequality in Eq. (22),

$$dU \leq T dS - P dV + \mu dN,$$

substitute to get

$$dF \leq T dS - P dV + \mu dN - T dS - S dT = -P dV + \mu dN - S dT.$$

At fixed temperature T , volume V , and particle number N ,

$$dT = dV = dN = 0,$$

we obtain

$$dF \leq 0.$$

Interpretation: Under these constraints, spontaneous processes decrease or keep constant the Helmholtz free energy. Equilibrium minimizes F :

$$\boxed{F \geq F_{\text{eq}}}$$

4. Enthalpy $H(S, P, N) = U + PV$ Differentiating,

$$dH = dU + P dV + V dP.$$

Using the inequality Eq. (22),

$$dU \leq T dS - P dV + \mu dN,$$

we get

$$dH \leq T dS - P dV + \mu dN + P dV + V dP = T dS + \mu dN + V dP.$$

At fixed entropy S , pressure P , and particle number N

$$dS = dP = dN = 0,$$

we obtain

$$dH \leq 0.$$

Interpretation: Spontaneous changes at fixed (S, P, N) decrease or keep constant the enthalpy, which attains its minimum at equilibrium:

$$\boxed{H \geq H_{\text{eq}}}$$

5. Gibbs Free Energy $G(T, P, N) = U - TS + PV$ Differentiating,

$$dG = dU - T dS - S dT + P dV + V dP.$$

Using the inequality Eq. (22),

$$dU \leq T dS - P dV + \mu dN,$$

substitute to find

$$dG \leq T dS - P dV + \mu dN - T dS - S dT + P dV + V dP,$$

which simplifies to

$$dG \leq -S dT + \mu dN + V dP.$$

At fixed temperature T , pressure P , and particle number N

$$dT = dP = dN = 0,$$

we obtain

$$dG \leq 0$$

Interpretation: Under these conditions, the Gibbs free energy decreases spontaneously and reaches a minimum at equilibrium:

$$\boxed{G \geq G_{\text{eq}}}$$

6. Grand potential $\Phi_G(T, V, \mu) = U - TS - \mu N$ Differentiating,

$$d\Phi_G = dU - T dS - S dT - \mu dN - N d\mu.$$

Using the fundamental inequality

$$dU \leq T dS - P dV + \mu dN,$$

we substitute for dU to obtain

$$\begin{aligned} d\Phi_G &\leq (T dS - P dV + \mu dN) - T dS - S dT - \mu dN - N d\mu \\ &= -P dV - S dT - N d\mu. \end{aligned}$$

Hence

$$d\Phi_G \leq -S dT - P dV - N d\mu$$

with equality for reversible processes. In particular, for processes at fixed temperature, volume, and chemical potential,

$$dT = dV = d\mu = 0 \quad \Rightarrow \quad d\Phi_G \leq 0.$$

Interpretation: Under these conditions, the grand potential decreases spontaneously and reaches a minimum at equilibrium:

$$\boxed{\Phi_G \geq \Phi_{G,\text{eq}}}$$

12.3 Summary

These extremum principles follow directly from the First and Second Laws of Thermodynamics. Under specified external constraints, equilibrium states are characterized by the extremization of the appropriate thermodynamic potential. In isolated adiabatic systems, equilibrium corresponds to a maximum of the entropy, and spontaneous processes occur with increasing entropy. For systems maintained at fixed temperature, pressure, volume, or other related constraints, spontaneous processes occur with a decrease of the corresponding thermodynamic potential, until its minimum value consistent with the constraints is reached.

Table 3: Thermodynamic Potentials, Fixed Variables, and Spontaneous Trends

Thermodynamic Potential	Fixed Variables / Process	Spontaneous Trend
Entropy S	Isolated, adiabatic	Increases to a maximum
Internal Energy U	S, V, N	Decreases to a minimum
Helmholtz Free Energy F	T, V, N	Decreases to a minimum
Enthalpy H	S, P, N	Decreases to a minimum
Gibbs Free Energy G	T, P, N	Decreases to a minimum
Grand Potential Φ_G	T, V, μ	Decreases to a minimum

12.4 Interpreting Thermodynamic Potentials out of Equilibrium

To understand the behavior of thermodynamic potentials in nonequilibrium situations, we take the Helmholtz free energy as an illustrative example. In equilibrium, F is a well-defined state function $F_{\text{eq}}(T, V, N)$ of its natural variables: temperature T , volume V , and particle number N . At fixed T, V, N — maintained by external control through a thermal reservoir, rigid walls, and a closed system — the equilibrium state is characterized by the variational principle

$$F_{\text{eq}}(T, V, N) = \min_{\substack{\text{all internal states} \\ \text{at fixed } T, V, N}} F,$$

where the minimum is taken over all microscopic arrangements compatible with the constraints.

When the system is out of equilibrium, global intensive quantities such as temperature and entropy lose their uniform, well-defined values. Under the *local-equilibrium approximation*, however, the system may be divided into volume elements each large enough to be considered locally homogeneous, yet small compared to the scale of inhomogeneities. In each element we can define a local temperature $T(\mathbf{r})$, entropy density $s(\mathbf{r})$, and internal energy density $u(\mathbf{r})$. The local free-energy density is then

$$f(\mathbf{r}) = u(\mathbf{r}) - T(\mathbf{r}) s(\mathbf{r}),$$

and the total free energy of a given nonequilibrium configuration becomes a *functional* of f :

$$F[f(\mathbf{r})] = \int_V f(\mathbf{r}) d^3r.$$

The central point is that the externally controlled parameters — the natural variables T, V, N — remain identical whether the system is in equilibrium or not. The inequality

$$F[f(\mathbf{r})] > F_{\text{eq}}(T, V, N)$$

therefore compares a specific nonequilibrium configuration (left-hand side) with the uniform equilibrium state that minimizes the functional (right-hand side), both under the same external constraints. The excess free energy $F - F_{\text{eq}}$ reflects the thermodynamic cost of the internal inhomogeneities and, as the system relaxes, provides the driving force for dissipation toward equilibrium.

This construction generalizes naturally to other thermodynamic potentials. In each case, the potential can be extended to a nonequilibrium functional via a local-equilibrium hypothesis, and its equilibrium value represents the extremum (minimum or maximum, according to the potential) of that functional under the externally fixed natural variables.

Appendices

A Proof that the Internal Energy of an Ideal Gas Depends Only on Temperature

For an ideal gas, the internal energy U depends solely on temperature T and is independent of volume V or pressure P . We prove this using thermodynamic relations.

Starting from the fundamental thermodynamic identity:

$$dU = TdS - PdV.$$

Express the total differential of U in terms of T and V :

$$dU = \left(\frac{\partial U}{\partial T} \right)_V dT + \left(\frac{\partial U}{\partial V} \right)_T dV. \quad (\text{A.1})$$

Similarly, write the differential of entropy S as

$$dS = \left(\frac{\partial S}{\partial T} \right)_V dT + \left(\frac{\partial S}{\partial V} \right)_T dV.$$

Substituting into the thermodynamic identity,

$$dU = T \left(\frac{\partial S}{\partial T} \right)_V dT + T \left(\frac{\partial S}{\partial V} \right)_T dV - PdV = T \left(\frac{\partial S}{\partial T} \right)_V dT + \left[T \left(\frac{\partial S}{\partial V} \right)_T - P \right] dV$$

Comparing this result with Eq. (A.1), we identify:

$$\left(\frac{\partial U}{\partial T}\right)_V = T \left(\frac{\partial S}{\partial T}\right)_V,$$

$$\left(\frac{\partial U}{\partial V}\right)_T = T \left(\frac{\partial S}{\partial V}\right)_T - P.$$

Next, we invoke the Maxwell relation derived from the fundamental thermodynamic potentials,

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V.$$

For an ideal gas, the equation of state is

$$P = \frac{nRT}{V}.$$

Thus,

$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{nR}{V}.$$

Substituting into the expression for $\left(\frac{\partial U}{\partial V}\right)_T$,

$$\left(\frac{\partial U}{\partial V}\right)_T = T \frac{nR}{V} - P = T \frac{nR}{V} - \frac{nRT}{V} = 0.$$

Thus, the internal energy does not change with volume at constant temperature. Therefore, the internal energy of an ideal gas is a function of temperature only:

$$\boxed{U = U(T)}$$